15,283
members

Who are ACDC's members? Exploratory analysis for segmentation

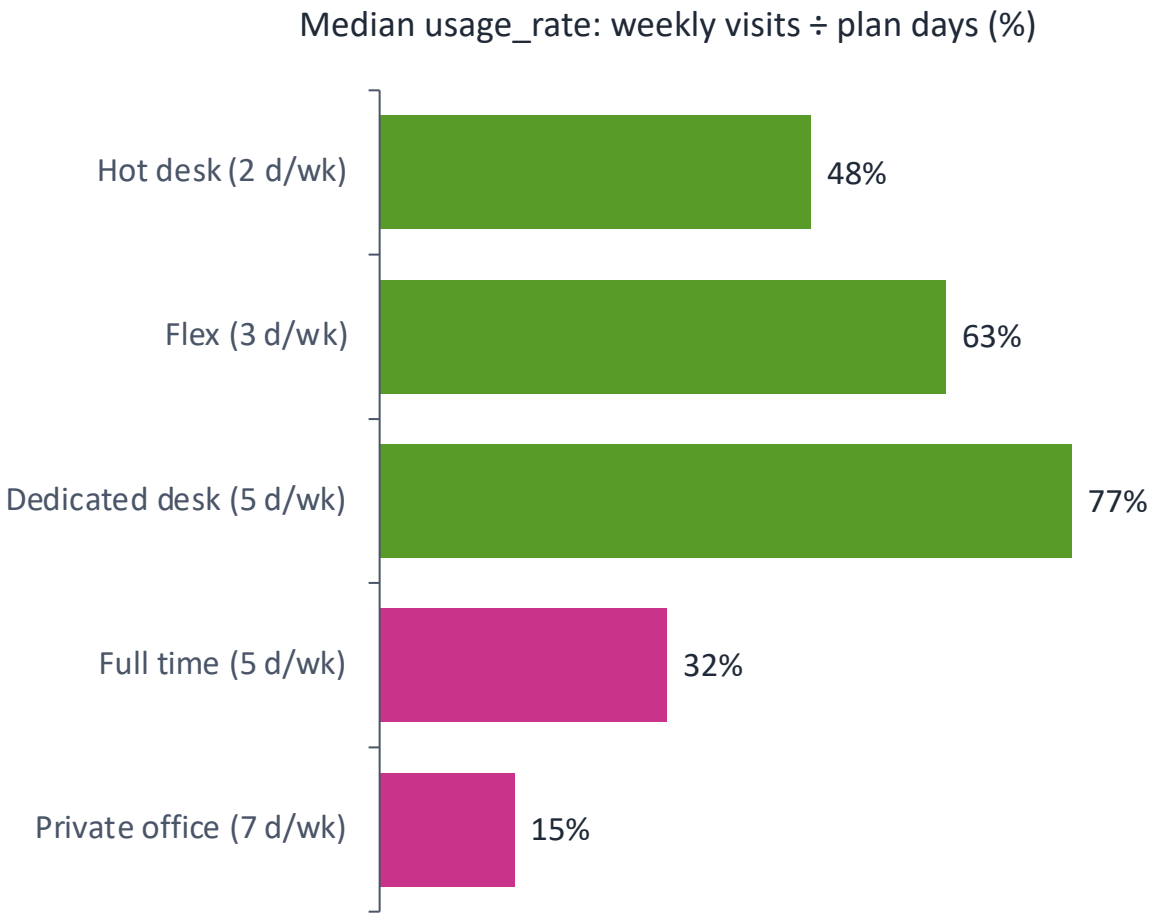
Group 34 - Data Mining 2026/27 - Nova IMS
Deliverable 1 - Exploratory Data Analysis - Fall Semester
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Five data issues could distort the clusters

Issue	Size of the problem	Risk for clustering	How to deal with it
Extreme and invalid lifetime value	Max €3.4M vs 99th percentile €25.7k; 1,720 zeros, 1,028 missing, 81 negative	A few members dominate distances and pull centroids	Use monthly_value and value_per_visit; log, cap at 1st/99th percentile, robust scaling
Placeholder recency	days_since_last_visit = 999 in 1,280 rows; 56 negative	Fake extreme absences	Recomputed from the last badge swipe
Inconsistent dates and renewals	149 memberships end before they start; 236 members with ≤ 1 month already renewed	Negative tenure; inflated renewal rates	Tenure set to missing; has_renewed used for profiling only
Last-period bias	Active members use 0.26 of their plan vs 0.60 for leavers	Clusters could split on status, not behaviour	Bias persists even in whole-membership usage: investigate before clustering
Impossible values	562 negative colleague counts, 96 plans outside 1–7 days, 50 members using > 100% of their plan	Outliers pull centroids away from real members	Set to missing or capped at 1

Already fixed in cleaning: 182 duplicate members removed; 1970-01-01 end date turned into an is_active flag (3,912 active); labels unified (gender 12 → 3, plan tier 19 → 5, country 33 → 8, industry 43 → 11); two date formats parsed.

Paying more doesn’t mean using more



Green - uses at least 40% of the plan
Pink - uses less than a third of the plan

91% vs 64%

Dedicated desk renews better than full time, and brings more per month (€393 vs €305): a fixed space matters more than the days included

€127 per visit

Private office: 4% of members and 19% of lifetime value, and 61% of them use less than a quarter of their plan

Same in every plan

Median age ≈ 34 and ≈ 28% team accounts in all tiers: demographics do not separate members

Engineered features capture value and behaviour

Perspective	Feature	Definition	Key number
Value	**tier_rank**	Plan tier as 1 (hot desk) to 5 (private office)	Replaces plan days (ρ = 0.98)
Value	**monthly_value**	Lifetime value / tenure (min. 1 month)	€128 hot desk → €697 private office
Value	**value_per_visit**	Lifetime value / total visits	€127 private office vs €25–31 others
Behaviour	**usage_rate**	Weekly visits / plan days per week	Median 0.52; 0.77 dedicated desk, 0.15 private office
Behaviour	**tenure_months**	Start date to end date (or 01-01-2024)	Median 8.2 months (left) vs 19.1 (active)
Behaviour	**recency_vs_usual_gap**	Days since last visit / usual gap between visits	Last visit a median of 2.3 usual gaps ago
Behaviour	**visit_trend**	Last-period visits / visits expected from own average	Median 0.99; 268 active members at 0
Profiling	**has_renewed · early_churn**	Renewed at least once · left within 3 months	28%–92% renewal by plan; 39% of leavers go early

Dropped as redundant (|ρ| > 0.8): plan_days_per_week, entitled_visits_last_period, visit_utilization_rate, avg_weekly_visits, n_renewals, lifetime_value.

Profiling only: age group, gender, country, industry, team account, is_active, has_renewed, early_churn.

We expect four to five member groups

69%

Casual flexible users

Hot desk and flex

Uses 48–63% of plan

28–46% renew

€128–195 / month

90% of early leavers

Improve onboarding

10%

Committed fixed-desk

Dedicated desk

Uses 77% of plan

91% renew

€393 / month

Best usage of any plan

Keep and reward

4%

Premium, rarely present

Private office

Uses 15% of plan

92% renew

€697 / month

€127 per visit; 19% of lifetime value

Protect: high value, low use

16%

Under-using full-timers

Full time

Uses 32% of plan

64% renew

€305 / month

Pays for 5 days, uses about 1.6 a week

Offer a better-fit plan

10%

Dormant members

Active, but no visit in 60+ days

1,513 members

39% of active members

May be a flag across groups

Re-engagement campaign

Expected from the EDA, not yet clustered. Circles show the share of members in each plan group; dormant members (green) overlap all four. We expect about 3–4 groups within each perspective and 4–5 in the final merged segmentation, chosen in Deliverable 2 with the elbow method, silhouette scores and dendrograms.

What this means for ACDC, in plain terms

How many groups?

4–5

Four plan-based groups are clear.

Dormant members may be a fifth group or a warning flag inside the others.

What separates them

1. How much of their plan they use
2. What they pay per month and per visit
3. How long they stay
4. Whether they renew (28% to 92% by plan)

Not age, gender, industry or team vs individual: these look the same in every plan.

Watch-outs

- Value data has extreme outliers (up to €3.4M) and many zeros: it must be capped and checked first.
- Active members appear to use their plan less than leavers: a data question to resolve before clustering.
- 1,513 paying members have not come in for more than two months: the first retention priority.

In short, ACDC's members differ less by who they are than by how they use and pay for their space: the segmentation should be built on value and behaviour.